

AUTUMN SEMESTER EXAMINATION-2021

Course Code: **PHYUGCC02**Course Title: **Mechanics**Department : PHYSICS, UG-1st Year, SEM-IFull Marks: **40**Time: **2 hrs****(Use separate answer sheet for each group)****Group-A****(Answer any FOUR questions, each having 5 marks, from the following)**

1. (a) Using the principle of conservation of energy, show that angular speed of a simple pendulum is

$$\dot{\theta} = \sqrt{\frac{2}{m l^2} [E - mgl(1 - \cos\theta)]}$$

where the symbols have their usual meanings.

- (b) A 6000 Kg rocket is set for vertical firing. If the gas exhaust speed is 1000 m/s, how much gas must be ejected per second to supply the thrust needed (i) to overcome the weight of the rocket and (ii) to give the rocket an initial upward acceleration of 20 m/s².
2. At time $t = 0$ a parachutist having weight mg is located at $z = 0$ and is travelling vertically downward with speed v_0 . If the force of air resistance acting on the parachute is proportional to the instantaneous speed, find the (a) speed, (b) distance traveled and (c) acceleration at any time $t > 0$. Find also the limiting speed of the parachutist.
3. (a) A particle follows a spiral orbit given by $r = c \theta^2$ (where c is a constant) under an unknown force field. Prove that such an orbit is possible in a central force field. Also find the form of the force field.
- (b) A planet of mass m and angular momentum L moves in a circular orbit in the a potential $V(r) = -\frac{K}{r}$ where K is a constant. If the planet is slightly deviated radially from it's path, find the frequency of it's radial oscillation.
4. Calculate the deviation from the vertical suffered by a body dropped from the rest at height h from the surface of the earth at the latitude λ (earth rotational velocity, ω) due to the Coriolis force.
5. Write down the postulates of Einstein's Special Theory of Relativity . Derive the Lorentz Transformation equation from it.
6. (a) A woman leaves the earth in a space craft that makes a round trip to the nearest star 4 light-years distant, at a speed of 0.9 c. How much younger is she upon her return than her twin sister who remained behind on the earth surface?
- (b) If the kinetic energy of the relativistic free particle is equal to its rest mass-energy, then find the velocity of the particle.

P.T.O.

Group-B

(Answer any TWO questions from the following)

7. (a) What is the axial modulus? Deduce the relation among the axial modulus, bulk modulus and rigidity modulus.
(b) What is flexural rigidity? A uniform beam is supported horizontally near the ends on two knife edges and a concentrated load W is applied at any point of the beam. Neglecting the weight of the beam, calculate the depression at any point of the beam.
(c) A wire is rigidly fixed at one end and a torque $1500 \times 10^{-7} \text{ Nm}$ applied at the free end produces shearing angle 60° . What will be the torque necessary for 1 radian angle of shear?
(1+3)+(1+3)+2
8. (a) What is an ideal fluid? Deduce Poiseuille's formula for the flow of liquid through a narrow horizontal tube.
(b) Two vessels of equal cross section ' α ' are joining near their bases by a horizontal narrow tube of length L and internal radius ' r '. Initially the liquid surfaces are at heights ' $3h$ ' and ' h ' respectively above the capillary tube. Calculate the time taken for the difference in level become ' h ' if the coefficient of viscosity is ' η ' and density is ' ρ '.
(c) Deduce an expression for velocity of efflux of a liquid using Bernoulli's theorem.
(1+4)+3+2
9. (a) Calculate the gravitational potential and intensity at a point on the axis of a uniform circular disc.
(b) Find the moment of inertia of a solid uniform right circular cone about an axis which is perpendicular to the axis of the cone and passes through its vertex.
(c) Show that the centre of oscillation and the centre of suspension for a given compound pendulum are interchangeable. What is the condition for the minimum time period of a compound pendulum?
3+3+(2+2)